

Scalable Community Mentorship: A Vision for Engineering Literacy & Access

Kyle Johnson^{*1,2} Vicente Arroyos^{*1,2} Liban Hussein^{*1,3} Aisha Cora^{*1,3} Eman Sherif^{*1,2} Celeste Garcia^{*1}
Teanna Barrett^{*1,2} Saara Uthmaan^{*1} Samira Shirazy^{*1,4} Jay Cunningham⁴ R. Benjamin Shapiro² Vikram Iyer²

Corresponding Author Email: chair@VELAccess.org

^{*} Equal and primary contribution first authors.

¹ AVELA - A Vision for Engineering Literacy & Access.

² University of Washington Paul G. Allen School of Computer Science & Engineering.

³ University of Washington Electrical & Computer Engineering.

⁴ University of Washington Human Centered Design & Engineering.

Abstract—Black and Hispanic/Latine communities remain underrepresented in engineering careers due to persistent economic, social justice, and human rights issues. Consequently, students from these communities face unique challenges when entering into engineering fields. Barriers stem from economic disenfranchisement, racial profiling, lack of community, and an absence of cultural competence. However, many programs that aim to tackle these issues are not holistic nor scalable to the degree needed for broader impact. To address the lack of holistic and scalable mentorship programs for Black and Hispanic/Latine students, we analyzed AVELA - A Vision for Engineering Literacy & Access' (AVELA) holistic student-led STEM engagement model. This model leverages multi-tier near-peer mentorship, mentor-embodied community representation, culturally responsive experiential & service-based learning, and compensated student-led community engagement. Thus far, AVELA has supported 361 university student instructors in teaching to 4,412 secondary school students across 213 classrooms. To evaluate members' motivations for participating in this mentorship program, we conducted 24 semi-structured interviews with undergraduate and graduate AVELA members. We find AVELA's holistic student-led STEM engagement model improves Black and Hispanic/Latine engineering students' experience in post-secondary education. Reduced power dynamics, community involvement, soft skill development, as well as financial support and technical skill building were indicated as key motivators for students' participation in the program. Students expressed that the depth, applicability, and scale of their mentorship relationships cultivated a reliable student community that empowered them to build a robust professional skill set.

Index Terms—Engineering Education, Underrepresented Minority Student, Mentorship Model, Participatory Design

I. INTRODUCTION

Black and Hispanic/Latine people are severely underrepresented in Science, Technology, Engineering and Mathematics (STEM), comprising only 14% of the U.S. engineering workforce and 19% of STEM bachelor degree holders, despite being more than 33% of the US population [1]–[3]. This low representation can be attributed to a myriad of reasons, such as economic disenfranchisement, racial profiling, lack of community, and the absence of cultural competence. These

factors create learning environments that exclude and/or antagonize Black and Hispanic/Latine students [4]–[7]. For the few Black and Hispanic/Latine students who do participate in engineering [8]–[10], they frequently report feeling unwelcome creating lower participation [11] and lower retention rates [12]–[14]. Therefore, to increase participation in STEM by Black and Hispanic/Latine students there needs to be more robust support systems that help alleviate some of these issues.

There are many prior works that have evaluated essential factors in improving engineering education for Black and Hispanic/Latine students. Approaches like assets-based community cultural wealth [15]–[17], transformative justice [18], [19], culturally responsive experiential and service learning [20]–[24], near-peer mentorship [25], [26], compensation [27], [28], as well as representative student-driven initiatives [9], [29], [30] have proven to benefit the quality of education for Black and Hispanic/Latine students because they help address systemic barriers and foster an inclusive, supportive, and empowering learning environment. Nevertheless, these isolated approaches fall short of reaching broader underrepresented minority (URM) communities because they do not simultaneously address the numerous barriers to access Black and Hispanic/Latine students experience when entering into engineering fields. Additionally, prior studies have shown the impact of representative mentorship in reducing racial profiling and increasing first year success [5], [10], [23], [31], however there are inherent limitations with scaling these programs given the relatively few Black and Hispanic/Latine professionals who can serve as mentors. All things considered, it is still unclear how URM mentorship programs should combine community, soft skill, and technical skill building in an assets-based manner to optimize student self-efficacy, belonging, and program scalability.

This work seeks to understand the motivations of AVELA - A Vision for Engineering Literacy & Access' (AVELA) Black and Hispanic/Latine mentors, as it pertains to supporting student self-efficacy and sense of belonging [32]. AVELA's nonprofit mission is to reduce access inequalities for Black

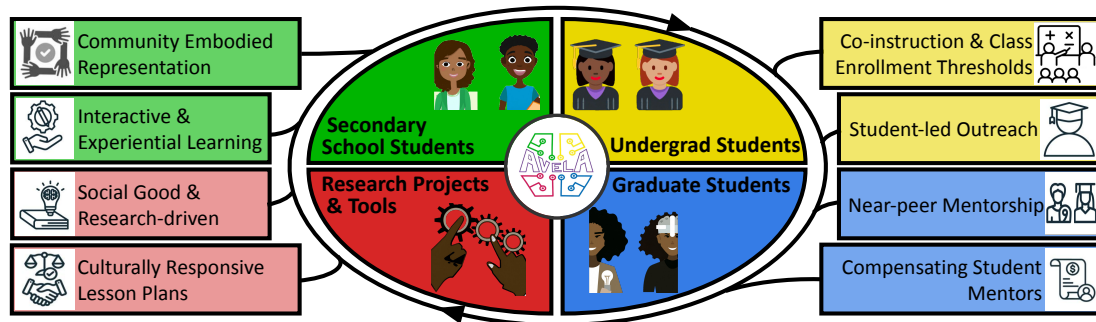


Fig. 1. Visualization of AVELA's holistic student-led STEM engagement model.

and Hispanic/Latine communities using a combination of multi-tier near-peer mentorship, mentor-embodied community representation, culturally responsive lessons plans, experiential and service-based engineering workshops, and compensated student-led community engagement approaches [32]. AVELA's multi-tier near-peer mentorship approach supports Black and Hispanic/Latine graduate students, with high-levels of technology and mentorship experience, in mentoring undergraduate students through the creation of experiential or service-based lesson plans, and the subsequent teaching of those lessons to secondary school students from similar communities. We refer to this specific combination of methodologies as a student-led STEM engagement model, depicted in Fig 1. Over the last 4 years, AVELA mentors have used this model to engage 361 university students from URM backgrounds in teaching engineering courses across 213 classrooms and community centers, providing 75,000 teaching hours to 4,412 secondary school students. AVELA's model has only succeeded at this scale because of highly engaged and committed student mentors. To better understand these mentors' motivations for participating in AVELA, we conducted and analyzed 24 semi-structured interviews with AVELA student mentors and presented our own reflections as founders, members, and supporters of AVELA's mission.

This study makes the following specific contributions towards understanding the impact of AVELA's holistic student-led STEM engagement model on Black and Hispanic/Latine mentors in the program:

- **Analyze the holistic student-led STEM engagement model** outlining 3 key design principles: A) Multi-tier near-peer mentorship and mentor-embodied community representation. B) Culturally responsive experiential & service-based learning. C) Compensated student-led community engagement.
- **Evaluate AVELA student motivations** for participating in the program. We find reduced power dynamics through less formal mentorship help improve Black and Hispanic/Latine students' self-efficacy and sense of belonging. We also affirm that giving back to the community, having spaces for building personal community, and soft skill development are key when designing mentorship programs for Black and Hispanic/Latine students.
- **Provide programmatic suggestions for academic**

stakeholders to consider when designing scalable mentorship programs for Black and Hispanic/Latine students.

In this study, we analyze and evaluate AVELA's holistic student-led STEM engagement model, which combines numerous pedagogical and culturally responsive practices. We investigate members' motivations for joining the program and suggest mentorship principles for academic stakeholders, like university administrators, that positively impact Black and Hispanic/Latine students' self-efficacy and sense of belonging.

II. BACKGROUND AND RELATED WORKS

A. Assets-based Community Cultural Wealth as a Framework

The predominant argument for low minority participation in engineering blames minority youth and communities, on a deficit-basis, for what they lack [33]. In particular, when applying a deficit-based lens to Black and Hispanic/Latine student persistence and retention in predominately White engineering programs, the reality of institutional barriers in college environments are often over-shadowed [34]–[36]. We build on research that rejects this framing. Community-cultural wealth has foundational grounding in intersectionality and critical racial identity, which presents an assets-based lens to understanding the persistence of URM students in engineering programs [37]–[39]. Yosso (2005) have defined various forms of cultural capital used by students of colors to succeed in higher education based on community assets [39]. Samuelson et al. (2016) further identified four dominant types of capital in which African American and Hispanic/Latine students leveraged: aspirational, navigational, familial, and resistant capital. Their research examined how these types of capital contribute to URM student participation, sense of belonging, persistence, and self-efficacy in undergraduate engineering programs [37].

B. Mentorship Programs- A National and Local Context

There are a myriad of mentorship frameworks and models that aim to address inequities in engineering education. These include integrational co-design [40]–[43], leveraging family-school-community leadership to be more culturally responsive [17], [44], service learning [45], near-peer mentoring [25], [26], [46]–[48], in-school and out-of-school programs with role models [49], and representative mentorship [31], [50], [51]. These approaches provide promising frameworks for addressing various aspects of engineering access inequality, however disparities continue to subsist [6], [7], [52]. There

remains a need to implement and evaluate a more scalable and holistic program which amalgamates various proven methodologies to address the multitude of different lived experiences and specifically articulated needs of URM students.

A transformative justice approach creates alternative structures and co-designed community-based solutions that center those who have been historically excluded from educational institutions [18]. While this can help increase URM students' sense of belonging, it alone does not address students' need for community, soft skill, and technical skill building. Experiential and service-based learning programs promote self-efficacy by engaging students with community, solving real-world problems, and promoting civic responsibility [22]–[24]. Nonetheless, these programs present logistical, quality control, and safety challenges, as well as being resource intensive and difficult to scale. Near-peer mentorship interventions and mentor-embodied community representation have shown promise in promoting self-efficacy [11], [27], [33], [46], but bootstrapping representative mentors with technical know-how remains a challenge that limits scalability.

In the U.S., individual and family monetary needs can outweigh students' interest in pursuing engineering education [53], [54]. However, DiSalvo illustrates how educational programs that provide both financial need and subject matter interest can create new inroads to engineering for marginalized students [28], [55]. Still, even with quality control and assessment, cost remains a limiting factor for the scalability of student compensated programming. Financial incentives also have not been shown to promote students' sense of belonging in engineering fields [27], [56].

C. Representative Student-led Organizations

Student organizations like the National Society of Black Engineers (NSBE) and Society of Hispanic Professional Engineers (SHPE) help prepare Black and Hispanic/Latine students for engineering careers through student-led leadership, service learning, and mentor embodied community representation [8], [10]. Student organizations have been shown to promote social, economic, and cultural capital amongst aspiring engineers [9]. However, these student-led organizations are typically dependent on support from university staff to facilitate smooth transitions between graduating and incoming student leaders.

III. METHODS

To understand the impact of AVELA's student-led STEM engagement model, we investigated members' motivation to join the program through structured interviews with undergraduate and graduate AVELA mentors. We received Institutional Review Board approval prior to conducting this study.

A. Positionality

Our interview participants are entirely from Black and Hispanic/Latine populations, and our research team is primarily composed of similar demographics. Student contributors are from primarily lower to middle class backgrounds, and many are the first in their families to attend college to study STEM.

With regard to gender, our team is equally represented by both women and men. Additionally, the research team included the founding members of AVELA. Prior to this study, 8 members of the research team had taught an AVELA course and were also active members in the organization. The research team's motivation to conduct this research arose from the success of AVELA's mentorship model compared to others, and the strong community AVELA members created for themselves. 10 members of the research team are students of the same U.S. public university as the participants, which could bias our analysis in a U.S.-centric way.

B. Recruiting and Participants

We recruited students who were active members of AVELA and who had taught an AVELA class. We asked students to fill out a interest form to collect information about their involvement in AVELA, access to technology education, and their demographic background. Figure 2 shows demographic data for all interview participants.

C. Interview Procedure

To mitigate bias, one author who is not affiliated with AVELA conducted 24 structured interviews over Zoom during the summer of 2023. The interviews followed a structured format and included questions about students interest in higher education, community and cultural values, experience in AVELA, and how AVELA impacted their future career. Interview sessions lasted approximately 45 minutes, and each participant was compensated \$20 for their interview.

D. Relevant Interview Questions

- 1) What were your motivations for obtaining higher education?
- 2) What were your motivations for obtaining a degree in STEM?
- 3) How did your upbringing affect your access to certain technologies?
 - a) Were there any parts of your religion, socioeconomic status, culture, language, family ideals, etc. that affected your access to certain technologies?
- 4) Have you worked on a project with AVELA that you believe had an impact on your community?
 - a) Can you describe that impact, was it an opportunity/experience that you shared with others?
- 5) Since joining AVELA have you received any mentorship from another AVELA member?
 - a) Did you perceive this mentor as a peer, or someone near your level?
 - b) What was the mentorship experience like?
 - c) What did you like about this mentorship?
 - d) What did you not like about this mentorship?
 - e) Were you comfortable asking questions to this mentor? Why or why not?
- 6) Since joining AVELA have you received any technical support from another AVELA member?

Interview ID	Age	Gender	Race/Ethnicity	Pell Eligibility	First Generation	Access to technology programs*	Participation in technology programs*	Number of AVELA classes taught
P1Yellow	22	Male	Black	Yes	Yes	Yes	Yes	5+
P2Green	19	Female	Black	Yes	Yes	Yes	No	1
P3Violet	24	Female	Hispanic/Latine	Yes	No	Yes	Yes	1
P4Red	19	Female	Black	Yes	Yes	Yes	No	2
P5Pink	20	Female	Hispanic/Latine	Yes	Yes	Yes	No	4
P6Silver	25	Male	Black	Yes	No	Yes	No	5+
P7Sapphire	19	Female	Black	Yes	No	Yes	No	1
P8Grey	22	Female	Black	Yes	No	No	No	2
P9Orange	31	Female	Black	Yes	No	Yes	No	4
P10Lime	40	Male	Black	No	Yes	No	No	1
P11Brown	20	Female	Black	Yes	Yes	No	No	5+
P12Coral	19	Female	Black	Yes	No	No	No	5+
P13Black	21	Female	Black	Yes	Yes	No	No	3
P14Cyan	22	Female	Black	Yes	Yes	No	No	2
P15Olive	24	Male	Black	Yes	Yes	Yes	No	2
P16Gold	20	Female	Black	Yes	Yes	Yes	No	2
P17Rose	28	Male	Black	Yes	Yes	Yes	No	2
P18Crimson	19	Female	Black	Yes	Yes	Yes	No	5+
P19Ruby	27	Female	Black	No	No	Yes	No	5+
P20Bronze	19	Female	Black	Yes	Yes	Yes	No	2
P21Marmalade	24	Female	Black	No	No	Yes	No	2
P22Sepia	23	Male	Hispanic/Latine	Yes	Yes	Yes	Yes	5+
P23Magenta	19	Female	Black	Yes	No	No	No	1
P24Emerald	23	Male	Black	Yes	No	No	No	3

* In secondary school

Fig. 2. Overview of interview participants' demographic information and number of AVELA classes taught.

- 7) Why did you join AVELA?
 - a) What was appealing about this organization?
 - b) What was unappealing about this organization?
 - c) What were motivating factors? Family, money, community?
- 8) Have your career plans, either long term (major, dream job, etc.) or short term (summer internship, class schedule, etc.) changed as a result of your participation in AVELA?
- 9) Through AVELA have you been made aware of new technologies that you see yourself using either in the long term (major, dream job, etc.) or short term (summer internship, class schedule, etc.)?
 - a) Have you used any skills or knowledge acquired through participation in AVELA outside of teaching an AVELA class?
 - i) Which skills or knowledge?
 - ii) How did you use them?
- 10) Within the context of AVELA, can you tell me about a time when your experience with technology further ignited your interest in the tech field?
- 11) Within the context of AVELA, can you tell me about a time when your experience with technology decreased your interest in the tech field?
- 12) Has participating in AVELA ever been a burden?
 - a) Why or why not?
- 13) Is there anything else you would like to say about your experience in AVELA or technology access and education?

E. Analysis Plan

First, two authors read through each interview transcript and corrected errors to match the recordings. Additionally,

they created pseudonyms for each of the interview participants to anonymize their identities. The analysis team consisted of 6 authors and utilized inductive thematic analysis to analyze the data. First, each author read 8 transcripts and identified significant statements and codes that explained students motivations for participating in AVELA, creating codes for each of the statements. Second, the analysis team met and performed affinity diagramming where we grouped significant statements that were similar, created a code based on the theme of the statements, and defined the code. During this process the team decided some significant statements did not adequately explain students motivations and were not included in the final list of codes. Third, the analysis team created overall themes that would explain students' motivations.

These themes formed the basis for our results. During this round we did not have any disagreement because the team finalized the themes together. We conducted our analysis using Hammer and Berland's guidelines [57], which recommend themes as claims about the data. We did not collect agreement metrics like inter-rater reliability, instead we resolved disagreements through discussion.

IV. RESULTS

By frequency of mention, we found that the themes most highlighted across all of our structured interviews were the need to give back to the community, a personal need for community, the ability to develop soft skills for lifelong engineering careers, and the reduced power dynamics throughout the organization.

A. Giving Back to the Community

The most frequent motivation highlighted by AVELA mentors for participating in the program was an interest in **supporting their communities by working with middle and**

high school students in the Seattle metropolitan area. Additionally, many students detailed their previous experience in teaching and community service as another motivator for giving back. They noted fulfilling social connections with younger students, sense of accomplishment in facilitating successful lessons, and increased personal passion in learning about technology. One student shared how AVELA's student-led outreach program differentiated itself from others and led them to join the organization.

I wanted an opportunity to give back to the community. I've seen a couple of different programs at UW that have similar missions that never really panned out. AVELA in particular has always been very good at actually delivering. Like, I was actually onsite in Seattle metropolitan area, and I was actually working with students, and I did receive compensation for my work. Which is a lot more than I can say for other programs at UW. - Participant 3

Mentors also expressed interest in giving back to their community to address the material barriers to accessing technical skills and resources. Several study participants mentioned having to overcome those barriers themselves to pursue engineering. In addition to technical skills and resources, participants wanted to provide students with advice on how to pursue post-secondary education. Mentors expressed understanding that for students who are underrepresented in STEM, it takes a community of support to overcome barriers to access. One mentor expressed the importance of students of color contributing towards the overcoming of systemic barriers:

The work of organizations like AVELA is very important to improve access and develop skills in technology because one, it gives students of color an opportunity to contribute ... by contributing you begin to ... come face to face with the barriers that exist out there. So I think also giving that opportunity to students who come from other communities to go out there and contribute to the progress that AVELA has in place is very important because it also helps to address these barriers by utilizing students who may have faced them as well. - Participant 10

B. Personal Need for Community

Another key motivator for the mentors in this research study was **finding a representative community that allowed them to grow and feel appreciated in a welcoming environment.**

The support from peers who understand my challenges and experiences makes me feel like I'm not alone in this journey. We uplift each other. - Participant 10

Participant 10 continued to highlight the aspect of shared experiences within the community, and having peers who have come across the same challenges and experiences as them, helped strengthen their self-efficacy.

I think having someone who's been through the same experiences as you, who looks like you, and has

succeeded, makes you believe that you can do it too.

- Participant 10

Another student shared that this community also gives hope for themselves in being successful. They stated that seeing someone who looks like them, who has shared experiences with them, and who is already successful helps them in believing that they will be successful as well.

AVELA was a community, especially when I was beginning to get into tech, where I can see people like me who already are in tech. When I joined most of the people that were in the club were people already in their majors. So it was like a way for me to see people who've already reached where I wanted to be, and to always also get help and also look at opportunities where I can gain skills, which are very hard to find in tech fields. - Participant 16

Students shared how AVELA gave them opportunities that they weren't able to find elsewhere when they first started their journey in engineering. They highlighted the student community as a place to get help and find opportunities to gain technical skills.

C. Soft Skill Development for Lifelong Engineering Careers

The opportunity to **learn, gain, and develop skills in communication, pedagogy, project management, and other soft skills** was another key motivator for students. Being 100% led by URM minority students, AVELA mentors have to lead course development, instruction, project management, as well as communication with other partners and community organizations. Several mentors shared that this experience helped them prepare for the multifaceted challenges that often arise within engineering careers.

Explaining concepts to my mentees helped me understand them better myself. It reinforced my knowledge and made me more confident in my abilities. Mentoring helps me develop leadership and communication skills that are crucial for my own career growth. It's a win-win situation. - Participant 10

Student-led leadership programs provide students with opportunities to improve on the soft skills that are often underemphasized in traditional technical courses. Student mentors who take initiative in communicating with partners also gain experience in negotiating and fundraising, which is a unique but a valuable experience for future engineering leaders.

I learned communication, [and] a little division of communication would be persuasion. I learned how to persuade folks [to partner with AVELA] and I learned how to articulate certain things so that people can understand it better. For example, rather than just saying all these coding languages, I would make it more digestible for folks so that they're at ease [with partnering with us]. Another skill is body language, the way I present myself. I really learned a lot in AVELA. - Participant 2

AVELA's multi-tiered near-peer mentorship structure engages many current and prospective graduate students. Graduate mentors reported significant improvement in their ability to communicate high-level scientific topics to diverse audiences, refining their science-communication and writing skills.

For the projects that I was involved in, I think I learned how to break down what I do as a grad student to an audience that's not familiar with biochemical concepts. So, I think that's an important skill because you have to explain [your research] to your community so that they understand why what you're doing matters. - Participant 19

D. Reduced Power Dynamics

Another aspect of the AVELA model that interested participants was the **informal mentorship approach that reduced the stress of performing and need to conceal personal attributes**. As stated by Participant 3, being able to mentor middle school students improved their ability to break down complex concepts for various knowledge levels, which translated into their teaching assistant (TA) job at university. Teaching first to middle school students in their community allowed for them to teach in a comfortable environment, reducing pressure and boosting their confidence.

After I worked with students at Seattle metropolitan area [Middle School], I was a TA for Physics 122 [at UW]. I had been a TA before, but this time I was teaching a bunch of labs, a bunch of tutorial sections. And because I had experience working with the students at Seattle metropolitan area [Middle School] that hadn't necessarily seen the kinds of technology that we were working with, I had a better understanding of the new college students and their level of experience. I was able to more easily connect with my students, and I was also a more engaging instructor as a result. Which is a benefit that comes with teaching middle schoolers that I don't think is talked about often enough. - Participant 3

AVELA mentors are also assigned to be representative of similar cultural, linguistic, community, and other backgrounds. AVELA's relaxed multi-tier near-peer mentorship model also helped alleviate the stress typically associated with hierarchical mentor-mentee relationships.

I felt comfortable asking questions because, like I said, we are in the same situation. Also the fact that he's black, I'm black, we're from African countries. I feel like we share a lot in common. So there's that kind of trust in the idea that he knows what I'm going through. And I also trust his sense of maturity because, he's gone through the whole process of undergraduate and now he's at grad school. So it's a couple of, you know, factors that just show how similar our paths are. And I think that helps to build trust in that person. - Participant 19

The reduced power dynamics in the AVELA mentorship structure also builds trust amongst mentors and mentees, especially when the relationship is built from shared experienced rather than hierarchical structure. Participant 19 stated because their mentor shared their cultural background and aspirations, they trusted their mentor's support and perspective more.

E. Additional Themes

In addition to the 4 most frequently mentioned themes throughout the interviews, participants shared several other notable motivations that were key in their personal choice to participate in AVELA's mentorship program. These additional themes include: **financial support/compensation, technical growth, and career growth**.

Participants noted that the financial compensation they received for their work with AVELA allowed them to remain full-time students instead of picking up a separate job.

Yeah, I would say, AVELA does help me a lot financially. I don't have to take a job during the school year when I'm working with AVELA because they take care of that. - Participant 20

Compensation helped make students' work feel valued and appreciated, leading to continued interest. Participants were also motivated to participate in AVELA as a means to explore new technology interests and gain new technical skills.

Through AVELA I was able to get into my major and I also was able to learn some coding skills and I was able to teach Python while still learning Python. And I believe that gave me a solid foundation for my future Python classes, and I was able to help my community. - Participant 23

Finally, participants viewed AVELA as an environment to nurture their career through the support, advice, and network of their mentors and the larger AVELA community.

...so they mentored me and helped me in many ways. Like for example, right now [redacted mentor] is helping me get into a research lab at the moment. - Participant 7

These additional themes further support the notion that AVELA's holistic student-led STEM engagement model offers wide-ranging benefits to the students who participate.

V. DISCUSSION

The findings derived from our study provide greater insight into the unique benefits of AVELA's holistic student-led STEM engagement model. This mentorship program has proven to be a holistic, student-led, and scalable model that addresses multiple aspects of support for URM students in engineering fields. The insights generated from our study confirm the findings of existing studies of mentorship models, and presents the unique benefits of AVELA's mentorship approach.

Multi-Tier Near-Peer Mentorship & Mentor-Embodied Community Representation There are many mentorship programs in engineering that provide undergraduate students with mentors who may have profound power or influence in the

field. While some studies have evaluated the promise of near-peer models as an alternative [48], our respondents uncovered a key benefit of the near-peer mentorship model that has been sparingly explored by previous research: relational power dynamics. Staunch power dynamics can be intimidating to students with no prior introduction to the field, as it can increase their nervousness and reinforce their internalized ideas of imposter syndrome, which is the belief that they are not fit to be in the positions that they are in [58]. With AVELA's multi-tier near-peer mentorship structure, undergraduates are paired with graduate student mentors from the same background or community. Many of our study participants noted that hearing advice from someone with their same upbringing and values created a deeper bond. This relationship showed to increase mentees' trust in their mentor's advice and belief that the mentor is truly invested in their success. Our findings demonstrate how this mentorship model can not only address mentorship power dynamics but create a mentorship relationship that uniquely enriches the URM student experience.

As AVELA is completely led by URM students, participants have the unique ability to be surrounded by fellow Black and Hispanic/Latine professionals in engineering, which is highly uncommon in the field. For example, Black and Hispanic/Latine engineers only make up 14% of the engineering workforce despite being 28% of the total workforce in the U.S [2] and more than 33% of the total U.S. population [1]. Previous research highlights how being surrounded by one's community helps to boost their confidence and reinforces their sense of belonging in the field, which is crucial to the longevity of their careers [15]. The sense of belonging described by the participants in the study not only provided individual confidence, but also demonstrated an interest in supporting the success of community-focused organizations like AVELA. Many students got involved, and stayed throughout multiple years and projects because of their developed need to give back to their larger communities, while also creating a new community for themselves in fields where they are grossly underrepresented. Other participants shared that they were introduced to AVELA by other URM students, which has led to the accumulation of more than 400 student members in AVELA since its inception in 2019 (with ~90% becoming instructors). Additionally, the opportunity for graduate mentors to not only give back to their general community, as stated by Participant 19, but to also support undergraduate students in growing their confidence helped to reinforce their own career choices. We posit that the student-led structure of the organization was a key factor in creating a flourishing URM student community, enabling the program to sustainably scale through effective student social networks.

Culturally Responsive Experiential & Service-Based Learning Soft skills are increasingly prioritized by engineering companies where collaboration between disciplines and effective knowledge dissemination is paramount. Despite the clear demand for soft skills among engineers, research has identified a lack of proficiency amongst engineers [59]. Since engineering coursework does not typically emphasize

the attainment of soft skills, emerging engineers have to find external means to develop these non-technical skills. Based on our findings, we suggest that other outreach programs should give more agency to their students mentors to lead their organizations to address these soft-skill shortcomings.

In AVELA, students can participate as elected board members, teachers and event organizers to gain vital leadership skills [23], [29]. Being a completely URM student-run organization, students engage in communicating with partners, sponsors, and stakeholders; grant writing; content creation; teamwork; science communication; and teaching. These skills have shown to translate into students' ability to communicate technical information to diverse stakeholders, with AVELA members having gone on to work at companies like NASA, Microsoft, Amazon, Boeing, IBM, Nvidia, and Texas Instruments, as well as get into Ph.D. programs at the University of California Berkeley, the University of California Los Angeles, the University of Washington, Johns Hopkins, Stanford, Yale, and the University of Michigan. The findings from this study demonstrate that student-run mentorship organizations such as AVELA can sufficiently provide soft skill training through real-world collaborations in a culturally responsive manner. This structured development of soft skills in the real world has shown to provide a greater sense of accomplishment and self-efficacy among the student members in AVELA.

Compensated Student-Led Community Engagement Study participants deeply identified with local communities due to shared racial identities and other perceived similarities in lived experience such as limited access to engineering resources. Mentors actively sought out opportunities to share the engineering resources they gained access to, especially with the larger URM student communities they identified with.

Participants were eager to "plant seeds" in the minds of the next generation. Participants felt that it was important to expose younger students to technology early in their academic career. Some participants credit their eventual interest and success in technology to their experiences of early exposure. Some participants simply wanted to widen the younger students' possible visions of the future to include engineering. Other participants were very passionate about their engineering interests and wanted to share the technical skills they developed with students as early as possible. Prior work has shown the potential to give back to your community is one possible reason why underrepresented students continue to participate in computer science [60]. Our findings show that it is possible for students to start giving back to their communities early in their academic careers, which often leads to higher retention rates of URM students. For these post-secondary students in college, they expressed wanting to create a space of encouragement for younger students. Through their teachings, college mentors aimed to impress upon the younger students that they can achieve anything they set their mind to.

In our analysis, a sense of fulfillment was the deeper motivation associated with giving back to the community. Participants were looking for opportunities to productively apply their technical skills, career experiences, and passions

for community through service. Several students noted that they had more bandwidth to continue their community work at AVELA because they received compensation. AVELA's student-led structure nurtures agency among its student participants to steer the direction of the organization based on the needs of the members, which enabled the organization to grow from 5 to 400 members in less than five years. By having agency in the direction and management of the program, students reported feeling more connected and proud of their work, leading to continued participation.

Limitations As AVELA founders, members, and supporters, our research team had insider likeness biases and relationships with interview subjects that may shape our interpretations of the data. The interview participants are those who responded to an email sent to an AVELA listserv over the summer, creating selection bias toward more engaged members. While our initial results are promising, broadening recruitment in future studies would diversify perspectives, and including a comparison group could help clarify this model's specific impact. Lastly, this paper provides an in-depth, contextualized look at how equity concerns emerged and were addressed within one specific program, between the summer of 2019 and the summer of 2023 in Washington state, for a particular population. Therefore, findings from this study should be put into this context and not over-generalized.

VI. CONCLUSION

This study demonstrates the significant impact of AVELA's holistic student-led STEM engagement model on enhancing self-efficacy and a sense of belonging among Black and Hispanic/Latine students pursuing engineering fields. By integrating multi-tier near-peer mentorship, mentor-embodied community representation, culturally responsive experiential & service-based learning, and compensated student-led community engagement, AVELA's model simultaneously addresses a multitude of unique challenges faced by Black and Hispanic/Latine communities. We conducted semi-structured interviews with 24 undergraduate and graduate members, revealing that reduced power dynamics, financial support, and opportunities for community and skill-building were key motivators for participation. These findings underscore the importance of designing mentorship programs that are comprehensive and culturally responsive to foster an inclusive and supportive environment for URM students in engineering.

Future work should focus on further scaling this mentorship model to reach broader audiences, while maintaining program quality and values. Additionally, exploring the long-term impacts of such programs on students' academic and career trajectories would provide further insights into the benefits of holistic mentorship approaches. By adopting the programmatic suggestions provided in this study, academic stakeholders can create scalable mentorship programs that not only enhance self-efficacy and belonging among Black and Hispanic/Latine students, but also contribute to a more diverse and inclusive engineering workforce. AVELA's model highlights the potential for transformative change in addressing the systemic barriers

that have historically hindered the participation and retention of URM students in engineering disciplines.

REFERENCES

- [1] U. C. Bureau, "Quickfacts washington," 2022.
- [2] R. F. BRIAN KENNEDY and C. FUNK, "6 facts about america's stem workforce and those training for it," 2021.
- [3] N. C. for Science and E. Statistics, "Science and engineering degrees earned," 2022.
- [4] U. E. E. O. Commission, "Eeoc fiscal year 2022 agency financial report," 2022.
- [5] E. McGee and L. Bentley, "The troubled success of black women in stem," *Cognition and Instruction*, pp. 1–25, 08 2017.
- [6] J. B. SAM DEAN, "Why are black and latino people still kept out of the tech industry?," 2020.
- [7] C. Pipa Stevens, "Digital racial gap could 'render the country's minorities into an unemployment abyss,' says deutsche bank," 2020.
- [8] S. of Hispanic Professional Engineers, "2022 shpe national convention: First look exhibitor opportunities," 2022.
- [9] N. S. Ross and S. McGrade, "An exploration into the impacts of the national society of black engineers (nsbe) on student persistence," June 2016. <https://peer.asee.org/27280>.
- [10] N. S. of Black Engineers, "Advancing the black stem experience," 2021.
- [11] D. A. Patterson Silver Wolf, F. Taylor, E. Maguin, and A. Asher Black-Deer, "You are college material—you belong: An underrepresented minority student retention intervention without deception," *Journal of College Student Retention: Research, Theory & Practice*, vol. 23, no. 3, pp. 507–522, 2021.
- [12] J. A. Kamer and T. T. Ishitani, "First-year, nontraditional student retention at four-year institutions: How predictors of attrition vary across time," *Journal of College Student Retention: Research, Theory & Practice*, vol. 23, no. 3, pp. 560–579, 2021.
- [13] A. Sithole, E. T. Chiyaka, P. McCarthy, D. M. Mupinga, B. K. Bucklein, and J. Kibirige, "Student attraction, persistence and retention in stem programs: Successes and continuing challenges," *Higher Education Studies*, vol. 7, no. 1, pp. 46–59, 2017.
- [14] A. A. Bergerson, B. K. Hotchkins, and C. Furse, "Outreach and identity development: New perspectives on college student persistence," *Journal of College Student Retention: Research, Theory & Practice*, vol. 16, no. 2, pp. 165–185, 2014.
- [15] C. C. Samuelson and E. Litzler, "Community cultural wealth: An assets-based approach to persistence of engineering students of color," *Journal of Engineering Education*, vol. 105, no. 1, pp. 93–117, 2016.
- [16] G. Lusk, T. Walker, and H. Ferraro, *Community Cultural Wealth in the Community College: A Systematic Review of Latinx Student Engagement*. PhD thesis, 2020.
- [17] S. A. Washington and L. Johnson, "Toward culturally sustaining/revitalizing indigenous family-school-community leadership," *Frontiers in Education*, vol. 8, 2023.
- [18] S. Erete, K. Thomas, D. Nacu, J. Dickinson, N. Thompson, and N. Pinkard, "Applying a transformative justice approach to encourage the participation of black and latina girls in computing," *ACM Trans. Comput. Educ.*, vol. 21, oct 2021.
- [19] I. Chordia, "Leveraging transformative justice in organizing collective action towards community safety," CHI EA '22, (New York, NY, USA), Association for Computing Machinery, 2022.
- [20] J. A. W. Rosenow, S. White, Z. Foste, A. Barry, and C. L. Voy, *Culturally Responsive Teaching in Practice: Exploring English Language Arts Classrooms That Serve Ethnically, Linguistically, and Culturally Diverse Students*. PhD thesis, 2020.
- [21] G. Gay, "Preparing for culturally responsive teaching," *Journal of Teacher Education*, vol. 53, no. 2, pp. 106–116, 2002.
- [22] B. D. Rose, E. Hill, and J. Beck, *Service Learning and the Experiential Learning Cycle in Elementary School*. PhD thesis, 2020.
- [23] B. Hotchkins, N. Hotchkins, B. Bellot, and L. Parker, "First year college student success for black and other students of color: A village initiative at the university of utah," *Journal of Curriculum, Teaching, Learning and Leadership in Education*, vol. 4, no. 1, p. 28, 2019.
- [24] G. Young, D. B. Knight, and D. R. Simmons, "Co-curricular experiences link to nontechnical skill development for african-american engineers: Communication, teamwork, professionalism, lifelong learning, and reflective behavior skills," in *2014 IEEE Frontiers in Education Conference (FIE) Proceedings*, pp. 1–7, 2014.

- [25] C. Sun and J. Clarke-Midura, "Testing the efficacy of a near-peer mentoring model for recruiting youth into computer science," *Mentoring & Tutoring: Partnership in Learning*, vol. 30, no. 2, pp. 184–201, 2022.
- [26] B. W.-L. P. Heather Pon-Barry and A. S. John, "Expanding capacity and promoting inclusion in introductory computer science: a focus on near-peer mentor preparation and code review," *Computer Science Education*, vol. 27, no. 1, pp. 54–77, 2017.
- [27] J. Clarke-Midura, V. Allan, and K. Close, "Investigating the role of being a mentor as a way of increasing interest in cs," in *Proceedings of the 47th ACM Technical Symposium on Computing Science Education, SIGCSE '16*, (New York, NY, USA), p. 297–302, Association for Computing Machinery, 2016.
- [28] B. DiSalvo, M. Guzdial, A. Bruckman, and T. McKlin, "Saving face while geeking out: Video game testing as a justification for learning computer science," *Journal of the Learning Sciences*, vol. 23, no. 3, pp. 272–315, 2014.
- [29] N. A. Alias and J. E. Luanan, *Student-Driven Learning Strategies for the 21st Century Classroom*. Advances in Educational Technologies and Instructional Design (2326-8905), IGI Global, 2016.
- [30] C. Luederitz, M. Meyer, D. J. Abson, F. Gralla, D. J. Lang, A.-L. Rau, and H. von Wehrden, "Systematic student-driven literature reviews in sustainability science – an effective way to merge research and teaching," *Journal of Cleaner Production*, vol. 119, pp. 229–235, 2016.
- [31] C. B. Newman, "Rethinking race in student-faculty interactions and mentoring relationships with undergraduate african american engineering and computer science majors," *Journal of Women and Minorities in Science and Engineering*, vol. 21, no. 4, pp. 323–346, 2015.
- [32] K. Johnson, V. Arroyos, C. Garcia, L. Hussein, A. Cora, T. Melaku, J. L. Cunningham, R. B. Shapiro, and V. Iyer, "Avela - a vision for engineering literacy access: Understanding why technology alone is not enough," 2024.
- [33] S. M. Ovink and B. D. Veazey, "More than "getting us through:" a case study in cultural capital enrichment of underrepresented minority undergraduates," *Research in higher education*, vol. 52, pp. 370–394, 2011.
- [34] S. R. Harper, "An anti-deficit achievement framework for research on students of color in stem," *New directions for institutional research*, vol. 2010, no. 148, pp. 63–74, 2010.
- [35] D. K. Grabsch, L. L. Moore, M. Levesque, and T. Robinson, "Using community cultural wealth: An asset-based approach to persistence of on-campus black and latinx collegians," *Journal of College Student Retention: Research, Theory & Practice*, p. 15210251231192884, 2023.
- [36] D. R. Parnell, J. Wilson, K. T. Hicklin, and J. A. M. Waisome, "Engineering while black: Exploring the experiences of black university of florida undergraduate engineering students using photovoice," in *2023 ASEE Annual Conference & Exposition*, 2023.
- [37] C. C. Samuelson and E. Litzler, "Community cultural wealth: An assets-based approach to persistence of engineering students of color," *Journal of Engineering Education*, vol. 105, no. 1, pp. 93–117, 2016.
- [38] O. Villalpando and D. G. Solórzano, "The role of culture in college preparation programs: A review of the research literature," *Preparing for college: Nine elements of effective outreach*, pp. 13–28, 2005.
- [39] T. J. Yosso*, "Whose culture has capital? a critical race theory discussion of community cultural wealth," *Race ethnicity and education*, vol. 8, no. 1, pp. 69–91, 2005.
- [40] H. Kang and J. M. Nation, "Transforming science learning framework: Translating an equity commitment into action through co-design," *Journal of Science Teacher Education*, vol. 34, no. 6, pp. 667–687, 2023.
- [41] A. Mawasi, W. Penuel, A. Cortez, and A. McKoy, "they were learning from us as we were learning from them": perceived experiences in co-design process," *Mind, Culture, and Activity*, vol. 0, no. 0, pp. 1–18, 2023.
- [42] S. Zuniga-Ruiz and K. D. Gutiérrez, "Pláticas as feminista cultural practice and design methodology for being and becoming with mathematics," *International Journal of Qualitative Studies in Education*, vol. 0, no. 0, pp. 1–12, 2023.
- [43] M. Lachney, W. Babbitt, A. Bennett, and R. Eglash, "Generative computing: African-american cosmetology as a link between computing education and community wealth," *Interactive Learning Environments*, vol. 29, no. 7, pp. 1115–1135, 2021.
- [44] K. D. Gutiérrez and A. S. Jurow, "Social design experiments: Toward equity by design," *Journal of the Learning Sciences*, vol. 25, no. 4, pp. 565–598, 2016.
- [45] R. G. Bringle and J. A. Hatcher, "Implementing service learning in higher education," *The Journal of Higher Education*, vol. 67, no. 2, pp. 221–239, 1996.
- [46] M. J. Lee, "Increasing minority youths' participation in computing through near-peer mentorship," *J. Comput. Sci. Coll.*, vol. 35, p. 47–56, oct 2019.
- [47] K. Kramarczuk, M. Narayanasamy, A. Khan, J. Plane, and K. Atchison, "Exploring the relationship between undergraduate near-peer intersectional computing, mentoring, and instructor identities," in *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education V. 2, SIGCSE 2022*, (New York, NY, USA), p. 1093, Association for Computing Machinery, 2022.
- [48] J. Clarke-Midura, V. Allan, and K. Close, "Investigating the role of being a mentor as a way of increasing interest in cs," in *Proceedings of the 47th ACM Technical Symposium on Computing Science Education, SIGCSE '16*, (New York, NY, USA), p. 297–302, Association for Computing Machinery, 2016.
- [49] C. Chen, J. Rothwell, and pedrito maynard zhang, "In-school and/or out-of-school computer science learning influence on cs career interests, mediated by having role-models," in *AERA 2023 Workshop on Multimodal Literacy in OST Programs: Family and Community Ties*, 2023.
- [50] D. Goh, C. Ogan, M. Ahuja, S. C. Herring, and J. C. Robinson, "Being the same isn't enough: Impact of male and female mentors on computer self-efficacy of college students in it-related fields," *Journal of Educational Computing Research*, vol. 37, no. 1, pp. 19–40, 2007.
- [51] K. Trenshaw, D. Rushton, E. E. Miskiöglu, and P. Asare, "Work-in-progress: Emergent themes from" high impact" role model and mentor narratives," in *2020 IEEE Frontiers in Education Conference (FIE)*, pp. 1–4, IEEE, 2020.
- [52] S. E. Sanders, Cynthia K., "The digital divide is a human rights issue: Advancing social inclusion through social work advocacy," *Journal of Human Rights and Social Work*, vol. 6, no. 2, pp. 2365–1792, 2021.
- [53] M. E. Maroto, A. Snelling, and H. Linck, "Food insecurity among community college students: Prevalence and association with grade point average," *Community College Journal of Research and Practice*, vol. 39, no. 6, pp. 515–526, 2015.
- [54] M. R. Silva, W. L. Kleinert, A. V. Sheppard, K. A. Cantrell, D. J. Freeman-Coppadge, E. Tsoy, T. Roberts, and M. Pearrow, "The relationship between food security, housing stability, and school performance among college students in an urban university," *Journal of College Student Retention: Research, Theory & Practice*, vol. 19, no. 3, pp. 284–299, 2017.
- [55] B. DiSalvo and K. DesPortes, *Participatory Design for Value-Driven Learning*, p. 14. Routledge, 1st ed., 2017.
- [56] L. Perlmutter, J. Salac, and A. J. Ko, "'a field where you will be accepted': Belonging in student and ta interactions in post-secondary cs education," in *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1, ICER '23*, (New York, NY, USA), p. 356–370, Association for Computing Machinery, 2023.
- [57] D. Hammer and L. K. Berland, "Confusing claims for data: A critique of common practices for presenting qualitative research on learning," *Journal of the Learning Sciences*, vol. 23, no. 1, pp. 37–46, 2014.
- [58] A. Parkman, "The imposter phenomenon in higher education: Incidence and impact," *Journal of Higher Education Theory and Practice*, vol. 16, pp. 51–60, 01 2016.
- [59] M. Hirudayaraj, R. Baker, F. Baker, and M. Eastman, "Soft skills for entry-level engineers: What employers want," *Education Sciences*, vol. 11, no. 10, p. 641, 2021.
- [60] N. Jaumot-Pascual, K. DeerInWater, M. Ong, and C. B. Silva, "'i can do data for my people': experiences of giving back for native undergraduates in computing," *Cultural Studies of Science Education*, vol. 18, no. 3, pp. 879–909, 2023.